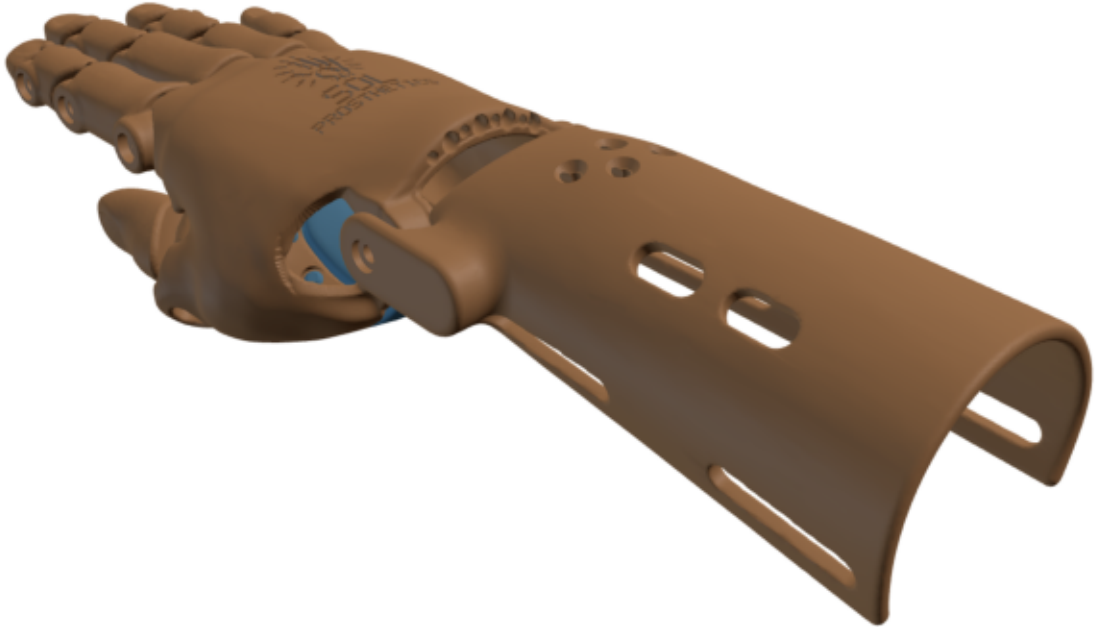
Bukang-liwayway v3.2

Illustrated printing, assembly and stringing guide

Use with the written Printing & Assembly Guide (parts list, print settings, tuning and care) and stl/HARDWARE.md for the size you print.

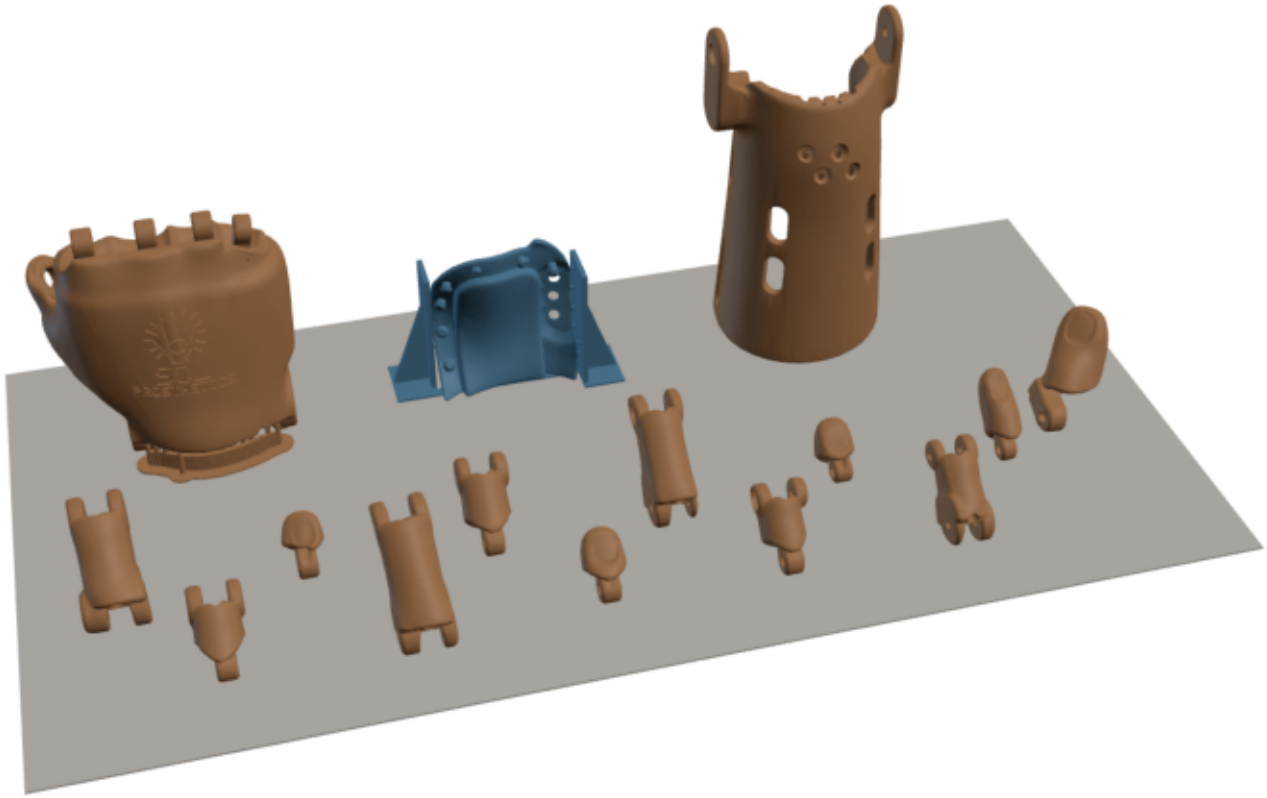


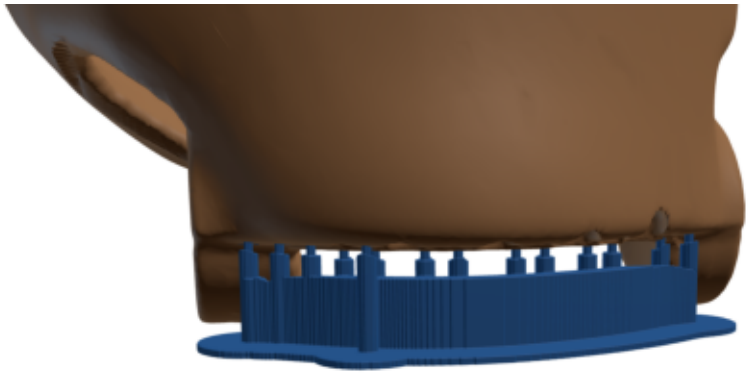


Order: print · 1 inserts · 2 fingers · 3 knuckle axle · 4 little finger · 5 thumb · 6 return cords
7 flexor lines · 8 wrist · 9 tensioner · 10 grip pads · 11 palm cover · 12 straps and liner

Printing — every part in its print direction

Load each STL as it is (never rotate or scale it). PETG = tan, TPU = blue. 0.4 mm nozzle, 0.2 mm layers.





palm_back_print.stl: the blue snip-off support holds the rim on 14 thin necks between the line-tunnel exits. Snip the necks flush after printing.

Loading the blocker

Creality Print / Orca / PrusaSlicer: drag both files in together, answer Yes to "multiple parts", right-click the blocker → Change type → Support blocker.

Cura: turn off "drop models to build plate" and "keep models apart", load both, blocker → Per Model Settings → Don't support overlaps.

Back of the hand

Print palm_back_print.stl standing on its wrist end. Normal supports ON, and load its support blocker file (..._support_blocker.stl) with it, so no support goes into a hole or line tunnel. 5 mm brim.

Fingers and thumb

They stand at 45°, knuckle end down, nail side up. Supports on the BUILD PLATE ONLY: that keeps every support out of the screw holes. Brim.

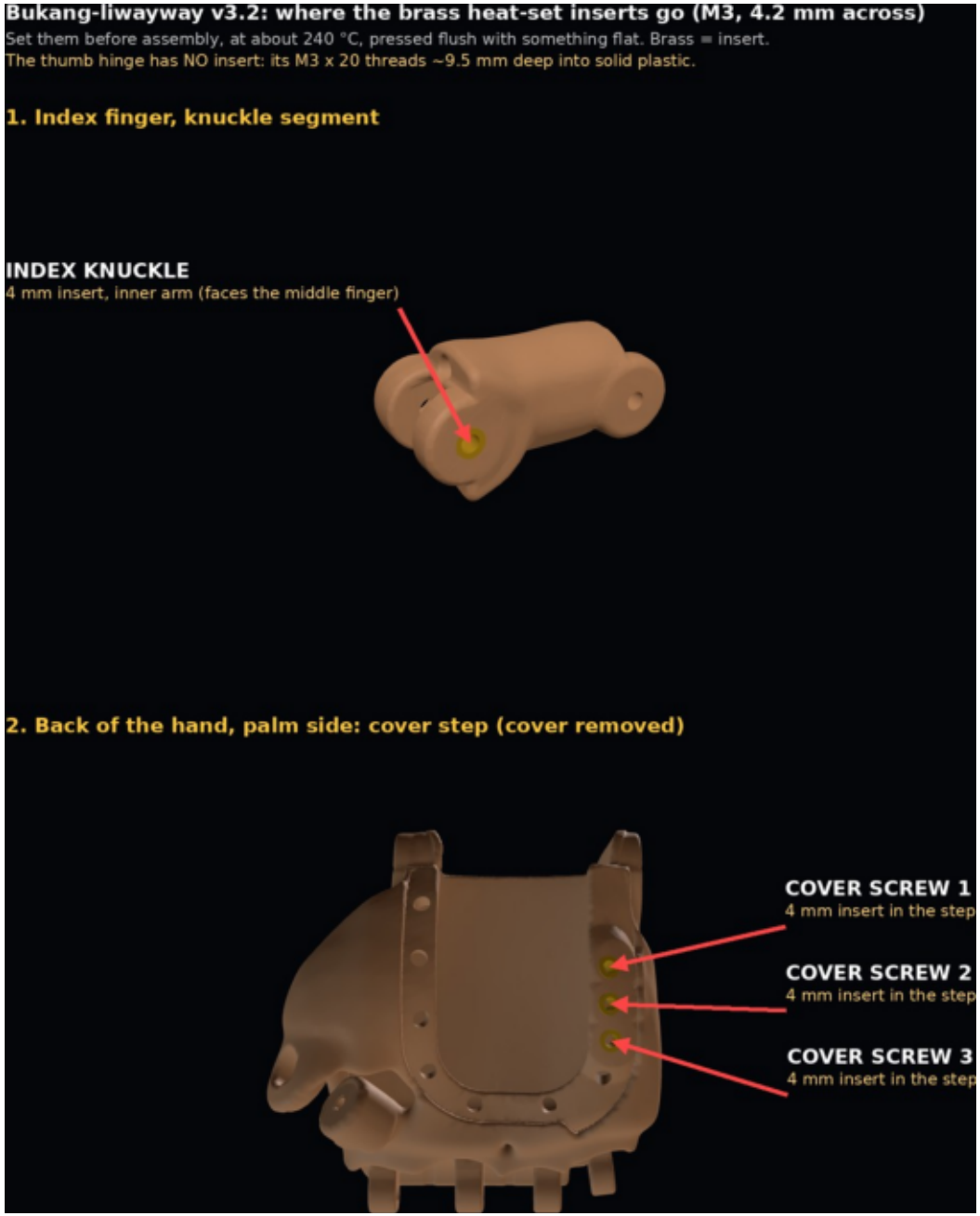
Palm cover (TPU 95A)

Print palm_flap_print.stl (fins and sole bar built in). Slow, 20–30 mm/s, 4 walls, little retraction, supports on the build plate only under the snap pegs, 5 mm brim.

Screw holes

Never need support: they all print lying sideways and are small (2.6–3.4 mm). Afterwards run a 3.5 mm drill through the 3.4 mm holes by hand, never the 2.6 mm pilots.

1 Heat-set inserts — set them first



Where

Four M3 brass inserts, 4.2 mm across × 4 mm long: one in the index knuckle segment's inner arm (the side that faces the middle finger) and three in the cover step along the little-finger edge of the back of the hand.

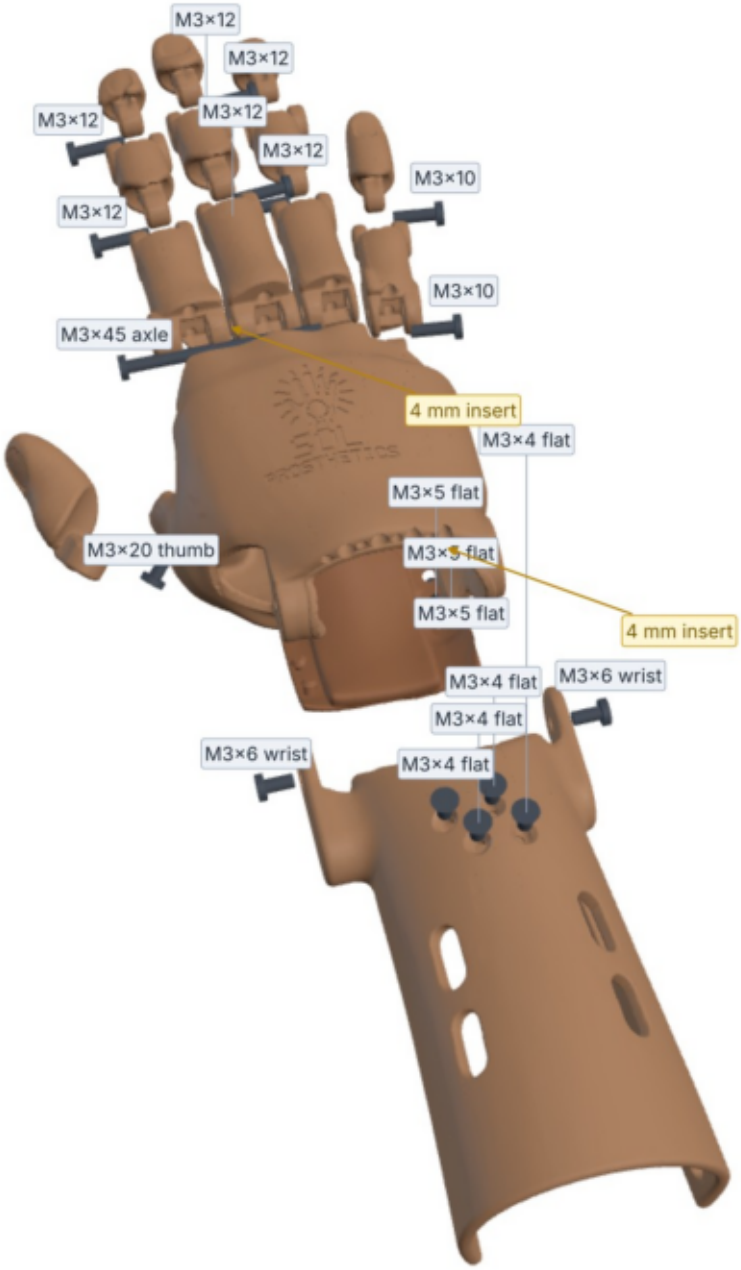
How

Soldering iron at about 240 °C for PETG (350 °C is too hot). Set the insert on its hole, let the iron's weight push it in, then press it flush with something flat and cool.

Not the thumb

The thumb hinge has no insert: its M3 × 20 threads about 10 mm into solid plastic.

Parts and screws



2 Build each finger

Join segment 1 to 2, then 2 to 3. Each joint is a barrel in a cup held by one button-head screw, put in from the side with the round recess so the head sits flush; it threads into the plastic, no nut. Index, middle, ring: M3 × 12 at both joints. Little: M3 × 10 at the tip joint.

3 Knuckle axle

Line up the index, middle and ring knuckles on their tongues. Push the M3 × 45 in from the index side through all three and screw it into the index insert until the head sits in its recess. It is a plain pin through middle and ring.

4 Little finger

Its own M3 × 10 from the recessed side.

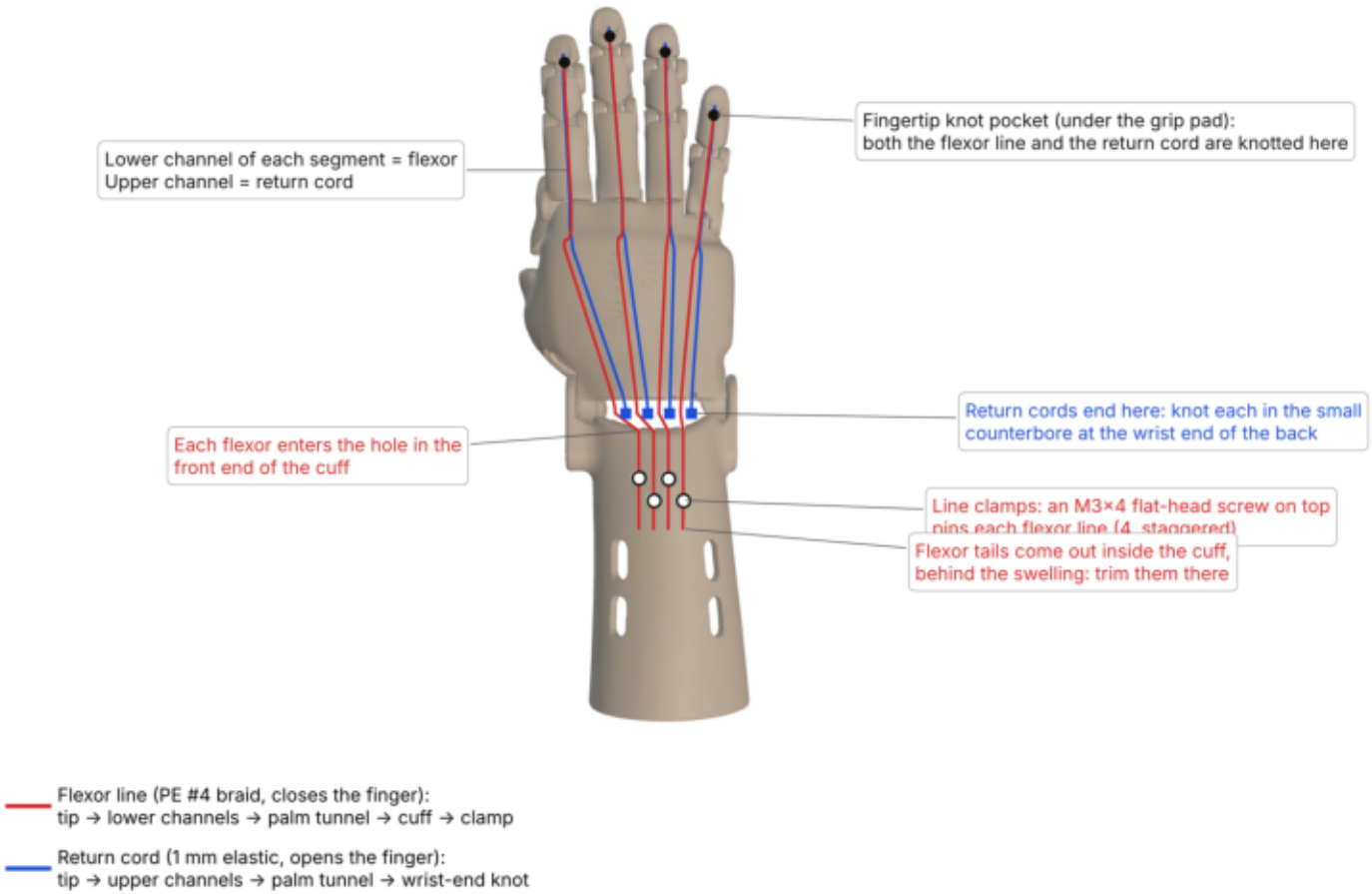
5 Thumb

An inner-tube washer on each side of the thumb tongue, into the slotted barrel, M3 × 20. Swing it to the grip and tighten until it holds.

Tighten

Only until each joint swings freely with no side wobble.

Stringing — the four fingers



6 Return cords (blue)

Knot the 1 mm elastic in the fingertip knot pocket (a surgeon's knot and a drop of glue). Thread it through the UPPER channel of each segment, then its tunnel in the back of the hand. Knot it in the small counterbore at the wrist end, just snug enough to hold the finger open.

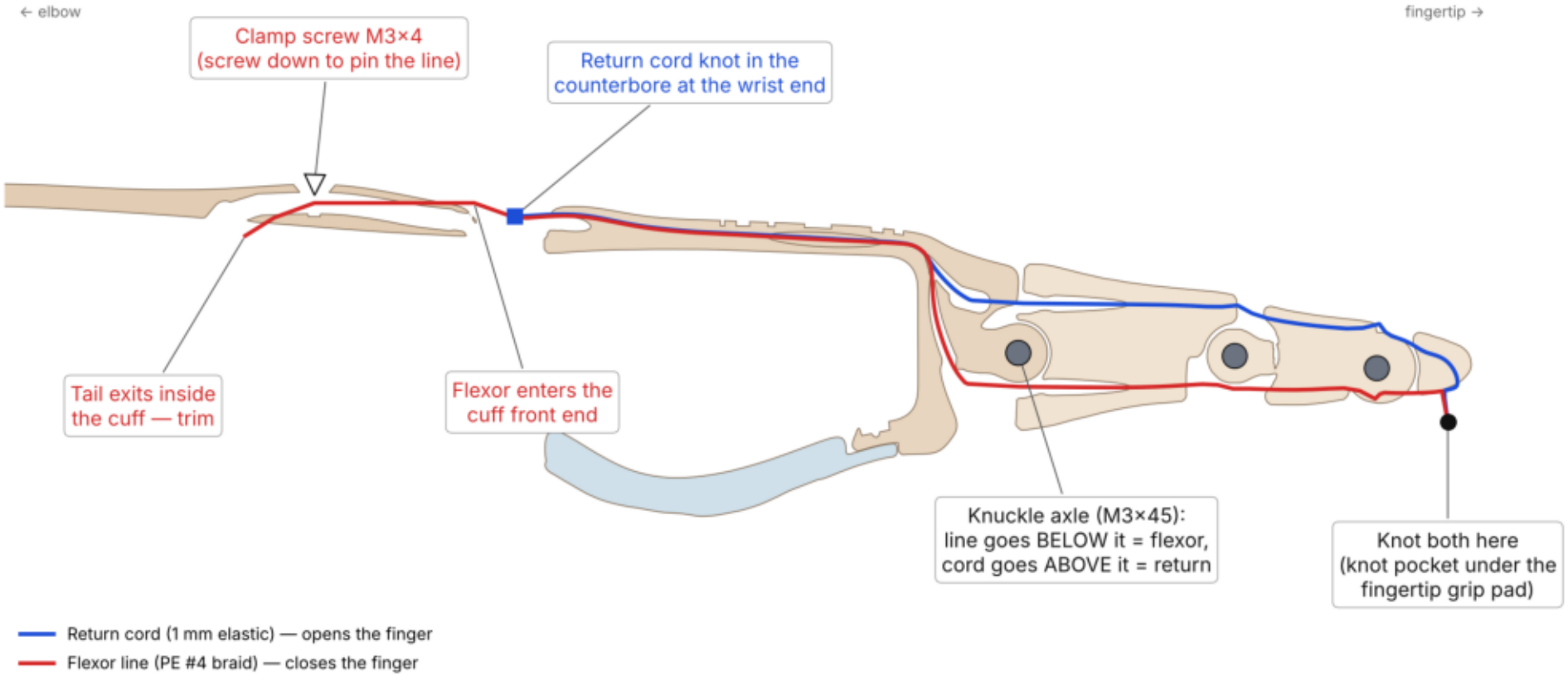
7 Flexor lines (red)

Knot the PE #4 braided line in the same fingertip pocket and seal the knot. Thread it through the LOWER channel of each segment, into the knuckle block and through its tunnel to the wrist end. Holes are 2–2.5 mm with 3 mm funnel mouths; a thin wire or long needle helps.

Tip

Do one finger completely before the next, and label the line ends with tape so you know which line belongs to which finger.

Stringing — side view of one finger (middle finger, cut lengthwise; the others are the same)



Which channel is which

Flexor = below the joint screws (palm side). Return = above them (back of the hand). Pulling the flexor bends every joint; the elastic springs them straight.

8 Wrist

Join the back of the hand to the cuff with two M3×6 screws through the cuff's hinge arms into the ear pads (the ear holes go right through).

9 Tensioner

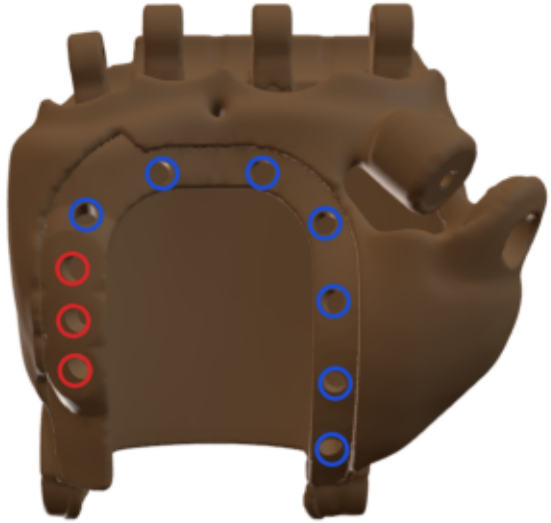
Feed each flexor into its hole in the front end of the cuff. With the wrist straight and fingers open, pull snug and screw down its M3×4 clamp until flush. Trim the tail inside the cuff. Bend the wrist down; adjust until all four fingers close together.

Grip pads, palm cover, straps

Palm cover (TPU), inside face



Back of the hand, palm side: the step



Red rings: the 3 hinge screws (M3 × 5 flat-head into the inserts), little-finger edge. Blue rings: the 7 push-through snap pegs and their holes in the step.

10 Grip pads

Inner-tube pads into the 5 fingertip and thumb recesses. Sand both dull, wipe with alcohol, thin film of E6000 or contact cement on both, wait until tacky (2–5 min), press, clamp 24 h. Cure 72 h before a child wears it.

11 Palm cover

Lay the hinge edge in the step and fit the three M3×5 flat-head screws into the inserts until flush. Don't overtighten. Fold the cover shut and press over each snap peg until it pops through (click). To open, peel the edge back firmly.

12 Straps and liner

Sew the two straps on through the cuff slots, fit the liner, sand any edge that touches skin. The child's hand goes in through the palm side and the open wrist end.